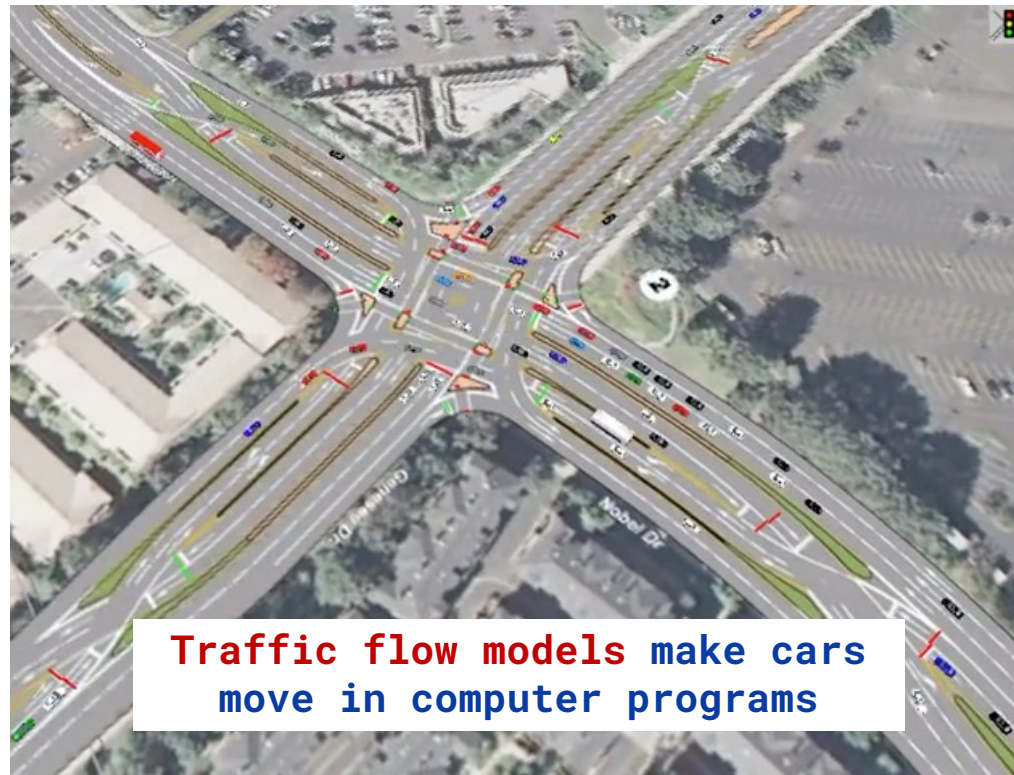


Cellular Automata & its Application in Traffic Simulation

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Traffic Simulation



Traffic flow models make cars
move in computer programs

Cellular Automata

One of the simplest traffic flow models

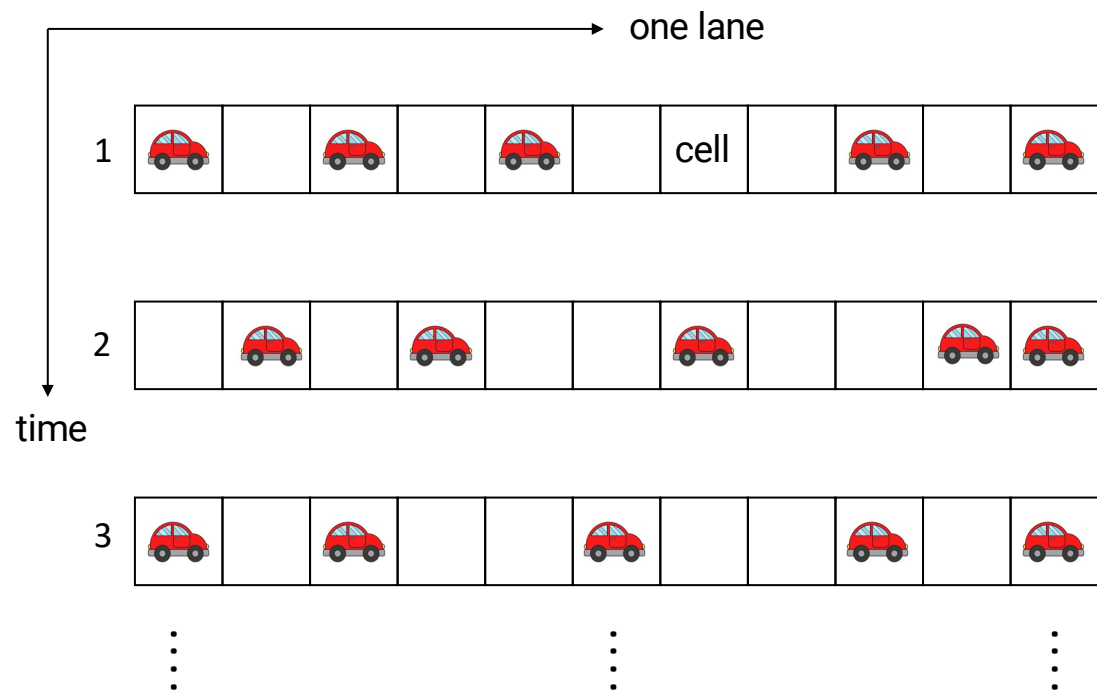
Glance

Cellular Automata in Traffic

- Lane is divided into cells
- Vehicles are in cells (position)
- Update vehicle positions
time by time
according to **rules**



Interaction Mechanism



longitudinal movement along a lane

Nagel-Schreckenberg Model

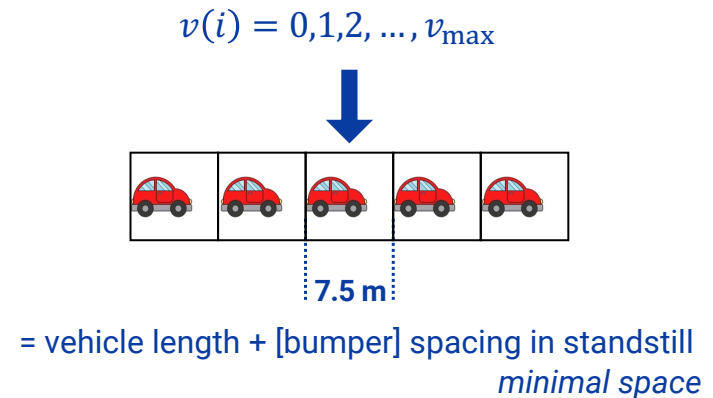
(NS/NaSch)

- Proposed by two German researchers K. Nagel and M. Schreckenberg in 1992
- Simplest and most generic/popular representative of traffic-related CA models

Reference: A cellular automaton model for freeway traffic. Journal de Physique I, 1992, 2, 2221–2229

NaSch Model: Setting

- A road is first divided into cells of length **7.5 m ?**
- Each cell can be either **empty** or **occupied** by **exactly 1 vehicle**
- Each vehicle is characterized by its current velocity $v(i) = 0, 1, 2, \dots, v_{\max}$ (**unit: cell**)
 $v_{\max}$ corresponds to a **speed limit** and is therefore the same for all vehicles (in the simplest case)
- The NaSch model simply consists of **4 Steps**:
Acceleration, Safety, Randomization, Driving

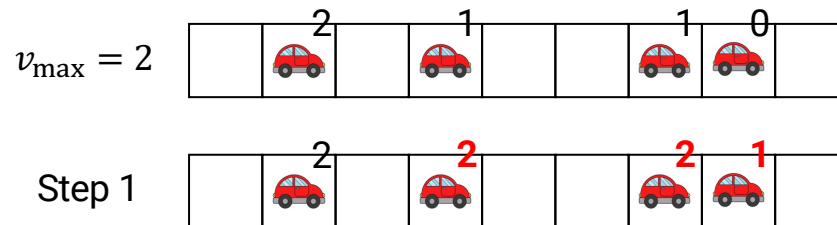


NaSch Model: Steps

Step-1: Acceleration

- Vehicle i , which has **not** reached the maximal velocity $v_{\max}$, accelerates by **one cell**

$$v(i) \rightarrow \min\{v(i) + 1, v_{\max}\}$$



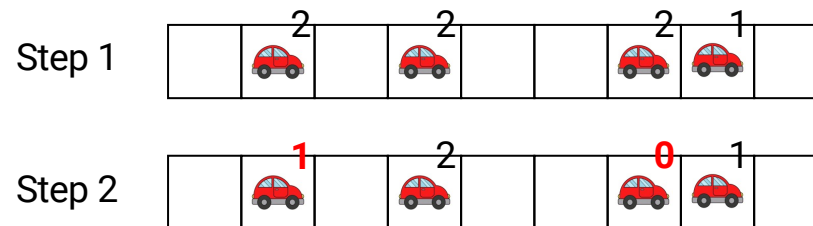
- This step describes **drivers' desire to drive as fast as possible** if allowed

NaSch Model: Steps

Step-2: Safety

- If $v(i)$ is larger than the number of its **empty cells ahead** $s(i)$, it reduces the velocity to $s(i)$

$$v(i) \rightarrow \min\{v(i), s(i)\}$$



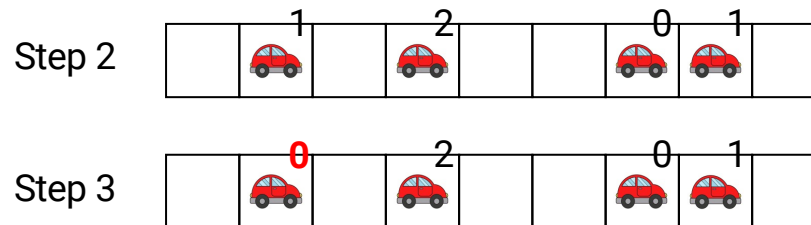
- This step encodes the **interaction between vehicles**. In this simple model, interactions only occur to avoid accidents (accident-free model)

NaSch Model: Steps

Step-3: Randomization

- With slowdown probability p , the velocity is **reduced** by **one cell**

$$v(i) \rightarrow \max\{v(i) - 1, 0\} \text{ with slowdown probability } p$$



- This step reflects **imperfect driving behavior**, and the simple microscopic setting results in many complex real-world macroscopic effects, such as traffic waves and **phantom traffic jams***

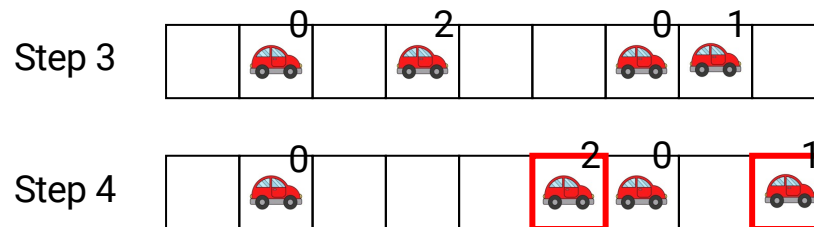
* Video Later

NaSch Model: Steps

Step-4: Driving

- **Physically** move vehicles forward to **new cells** according to **new velocity** $v(i)$ and last-time position $x(i)$

$$x(i) \rightarrow x(i) + v(i)$$



```
for veh in vehicles:  
    UpdateSpeed(veh)  
  
for veh in vehicles:  
    UpdatePosition(veh)
```

- Although we process ('move') the vehicles one-by-one, the actual movements are **PARALLEL** in computers, because all velocities have been previously determined

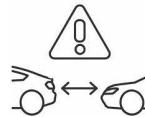
Summary

Minimal Model*: leaving out [one of the 4 steps] no longer leads to a realistic behavior

- Update Speed – **No vehicle moves**



Step 1
Driver desire



Step 2
Safety distance



Step 3
Imperfect behavior

Essential Parts

- Update Position – **Step 4** – Only step that **vehicle moves**



** capture the core features of a problem by simplifying it, improving interpretability and generalization.*

longitudinal movement along a lane

Velocity-Dependent-Randomization Model (VDR Model)

- In practice, the slowdown probability p is dependent on the vehicle velocity. While p in the NaSch model is constant.
- The VDR model incorporates the dependency by introducing a velocity dependent $p(v)$

VDR Model: Setting

Step-0: Determination of the randomization parameter $p(v)$

- For standing vehicles $p(v = 0) = p_2$
- For moving vehicles $p(v > 0) = p_1$
- Typically, $p_1 \ll p_2$, such as

$$p_1 = 0.01 \quad p_2 = 0.5$$

Questions?

Nagel-Schreckenberg Model

lateral movement across lanes

Lane-changing Model

Reference: Two lane traffic simulations using cellular automata. Physica A, 1996 4367(95)

Lane-changing

Step-1: Motivation (Acceleration)

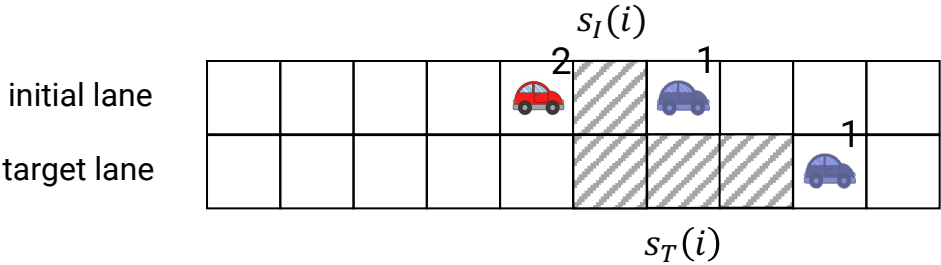
Step-2: Feasibility (Safety)

Step-3: Randomization

Step-4: Execution (Driving)

- A vehicle can drive **faster** by changing lanes

2. More space on the target lane : $s_T(i) > s_I(i)$, $s_T(i)$ – distance to front vehicle, **target lane**

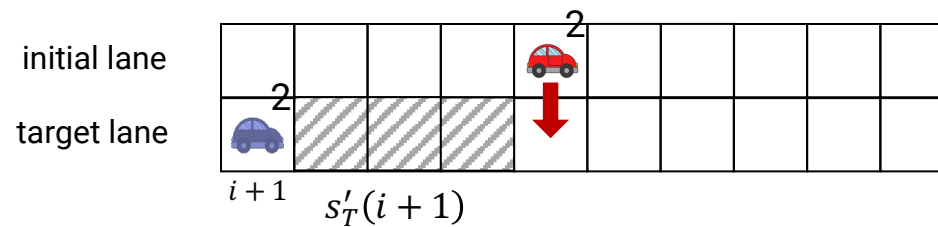


Lane-changing

Step-2: Feasibility (Safety)

- **No rear-end collision** if changing lanes

$$s'_T(i+1) > v_T(i+1)$$



- A vehicle is **not necessary** to change lanes even if the motivation and feasibility exist at the same time

- **Physically** process ('move') the vehicle to the target lane, which is **PARALLEL** in computers



Questions?

Lane-changing Model

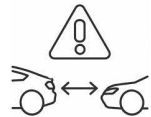
Phantom Traffic Jams: *Real World*



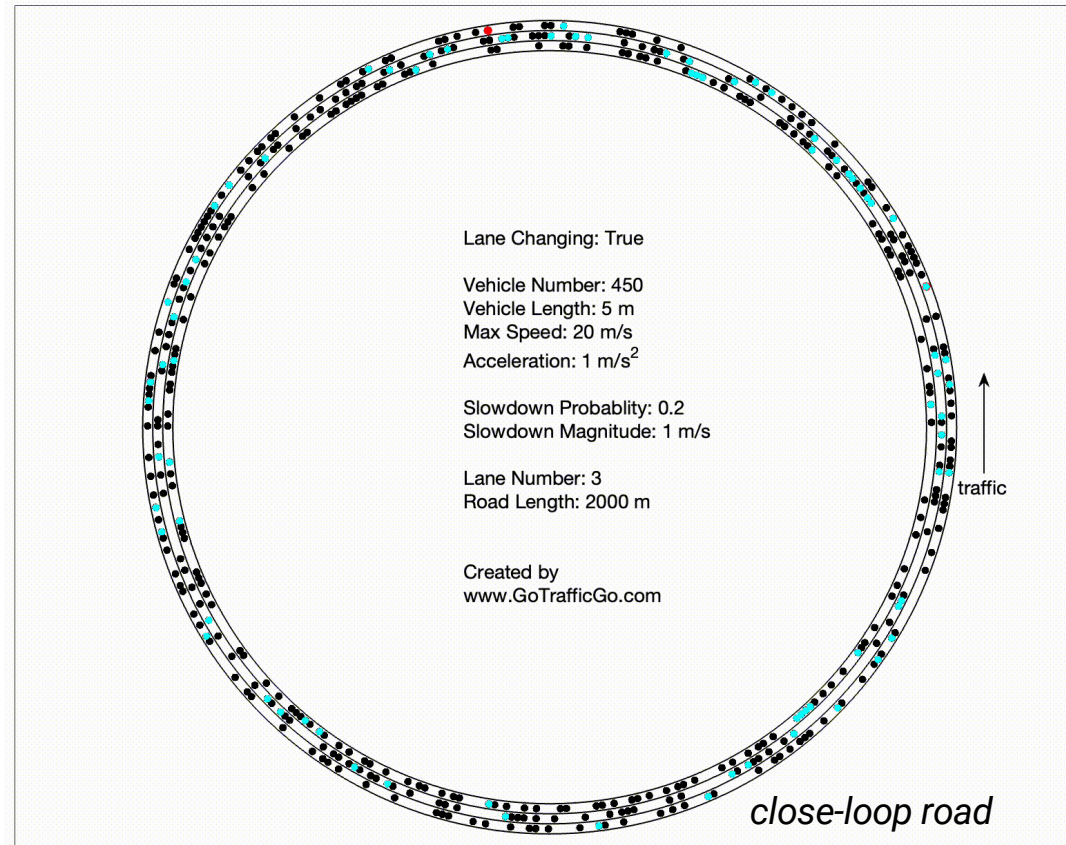
Phantom Traffic Jams: *Simulation*



Step 1
Driver desire



Step 2
Safety distance



Only difference: cell size = 1m (instead of 7.5m); a vehicle occupies 5 cells

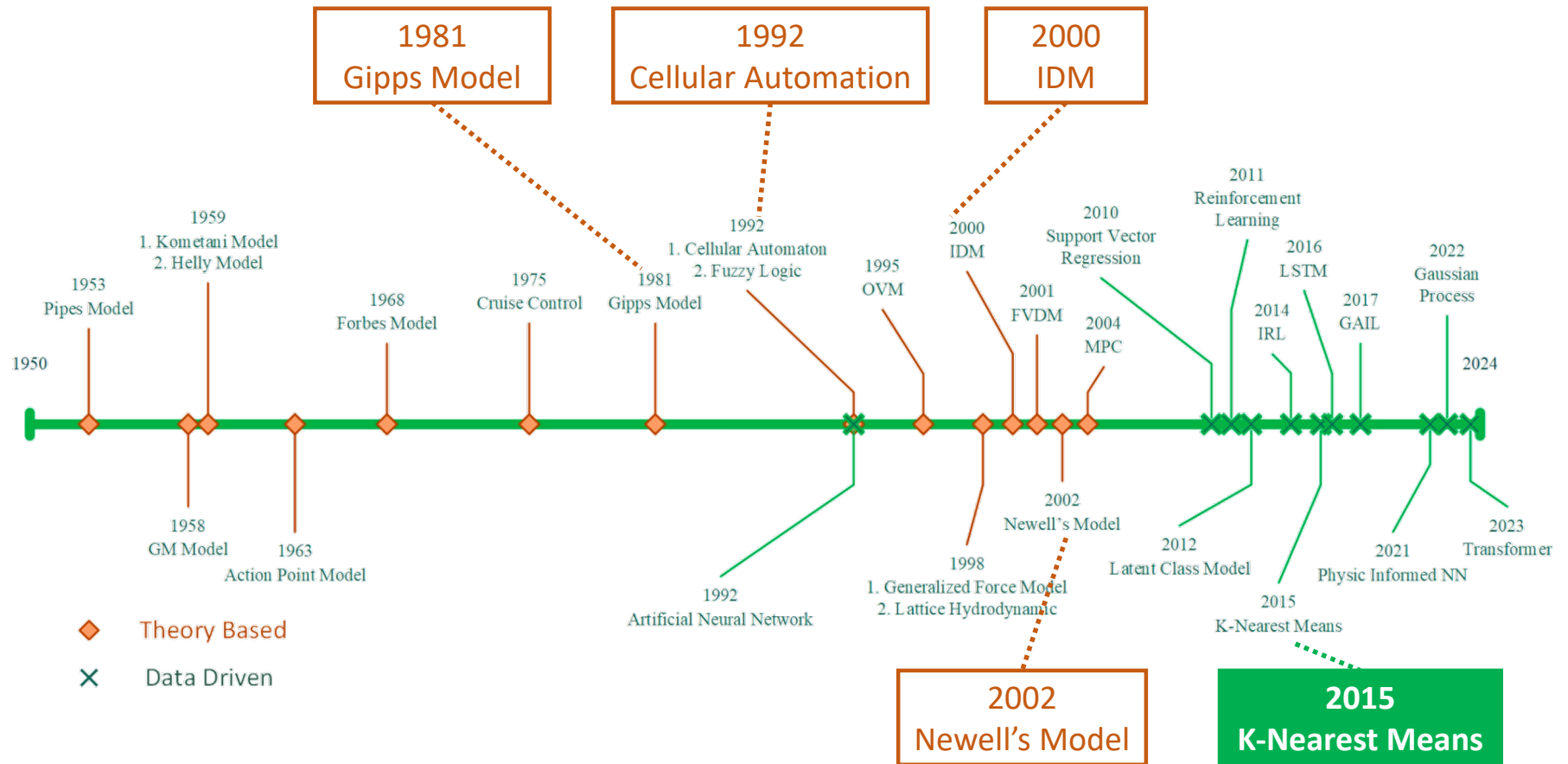


Step 3
Imperfect behavior

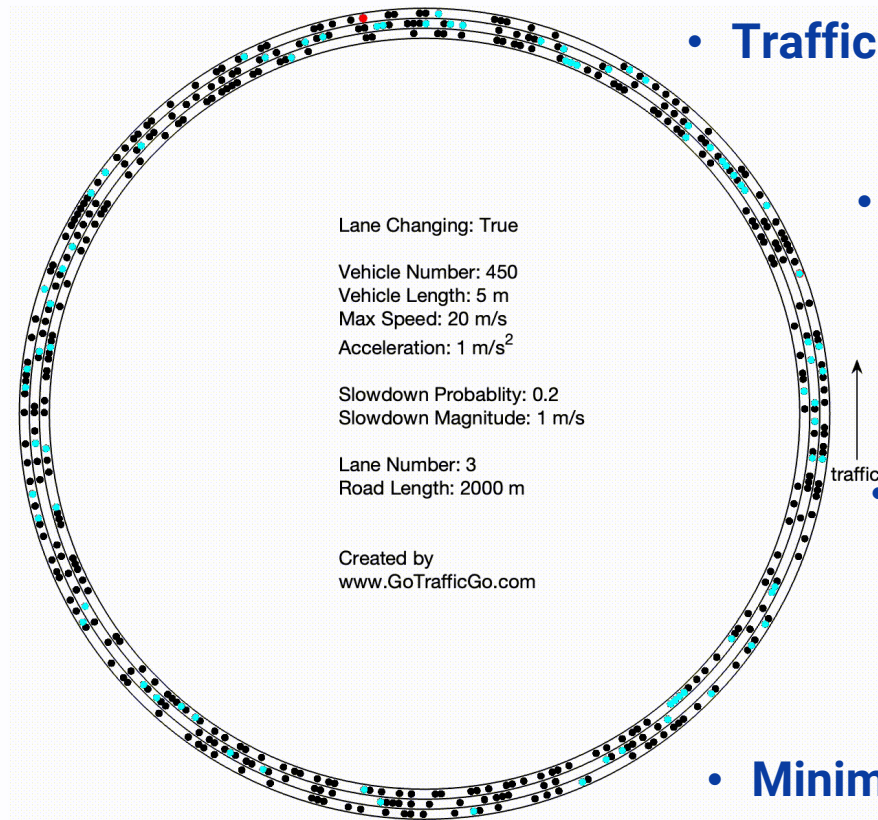


Step 4
Driving

Extension: Car-following Models



Takeaways: Direct Impression



- **Traffic Simulation**

- **Cellular Automata in Traffic**

- Longitudinal: NaSch Model
- Lateral : Lane-changing Model

- **One of the Causes of Traffic Congestion**

- Phantom Traffic Jam
- Please drive steadily

- **Minimal Model**

Thank You

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